IF THE SINGULAR IDEAL GENERATES THE UNIT IDEAL OVER LAMBDA THE MAP FACTORS THROUGH A SMOOTH ALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. The source ends 'Details omitted' for the concluding smoothness check of B; the check is carried as an explicit core obligation (invert f_i and eliminate x_i linearly; the f_i generate the unit ideal in B).

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 543 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Definition of the singular ideal. definition-strictly-standard (same chapter, structural). H_{A/R} is generated by elements a with A_a smooth over R.

Lemma 0.1. *Let $R \to A \to \Lambda$ be ring maps with A of finite presentation over R. Assume that $H_{A/R}\Lambda = \Lambda$. Then there exists a factorization $A \to B \to \Lambda$ with B smooth over R.*

Proof. Choose $f_1, \ldots, f_r \in H_{A/R}$ and $\lambda_1, \ldots, \lambda_r \in \Lambda$ such that $\sum f_i\lambda_i = 1$ in Λ. Set $B = A[x_1, \ldots, x_r]/(f_1x_1 + \ldots + f_rx_r - 1)$ and define $B \to \Lambda$ by mapping x_i to λ_i. To check that B is smooth over R use that A_{f_i} is smooth over R by definition of $H_{A/R}$ and that B_{f_i} is smooth over A_{f_i}. Details omitted. □

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.